Four datasets with the same statistics and four different stories

Why you should plot before you summarize: Anscombe's quartet (1973)

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All four sets share the same mean, variance, correlation and regression line

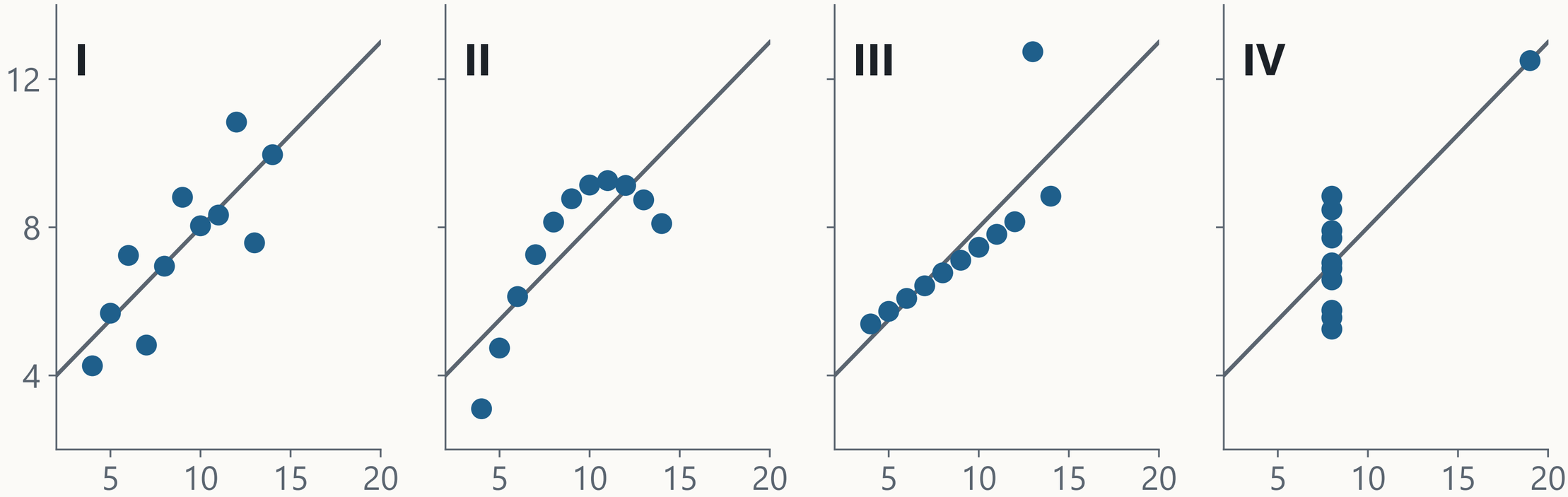
r = 0.816

In all four: mean of x **9**, variance of x **11**, mean of y **7.50**, variance of y **4.12** and fitted line **y = 3.00 + 0.500 x**.

If we only looked at this table, we would call them the same phenomenon.

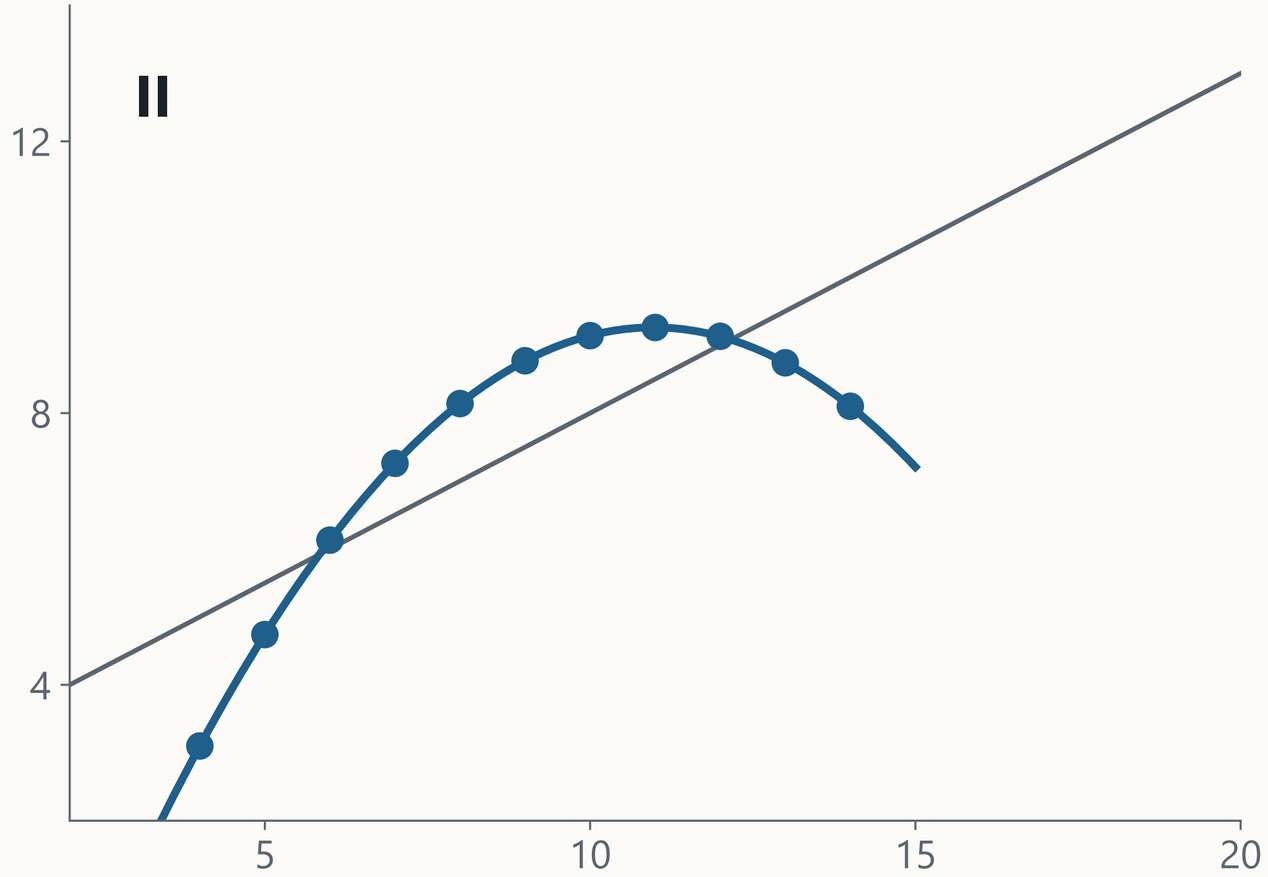
Source: Anscombe, F. J. (1973), The American Statistician 27(1), 17–21. Own computation in figuras/hacer_figuras.py.

Plotted, they are four different stories



Source: Anscombe (1973). The four panels share axes; the grey line is the same fit in all of them.

In II the data are a curve: the straight line is the wrong model

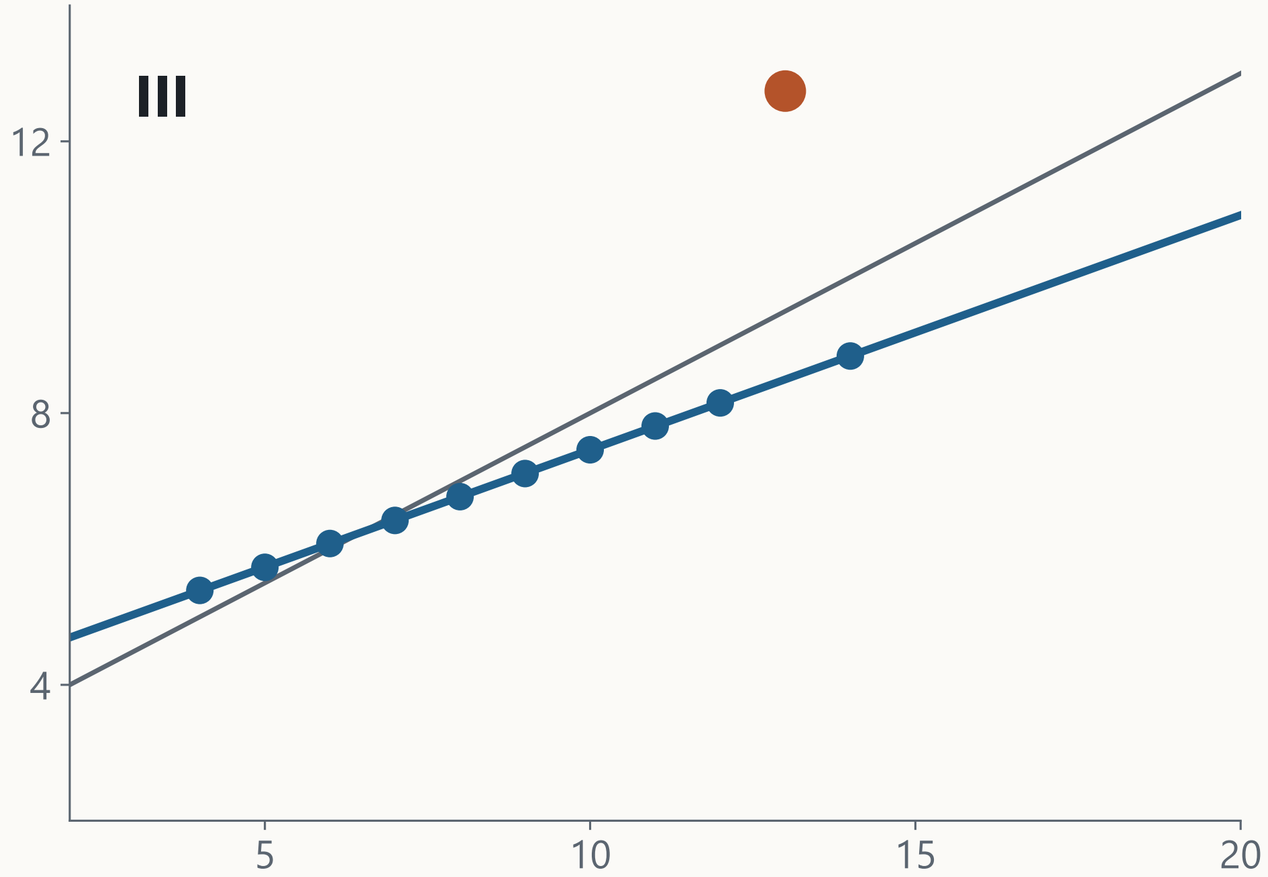


A parabola describes them almost exactly.

A correlation of 0.816 measures how well a line fits, not whether a line is the right model.

Source: Anscombe (1973); own quadratic fit.

In III a single point bends the line: without it, the data are perfectly aligned

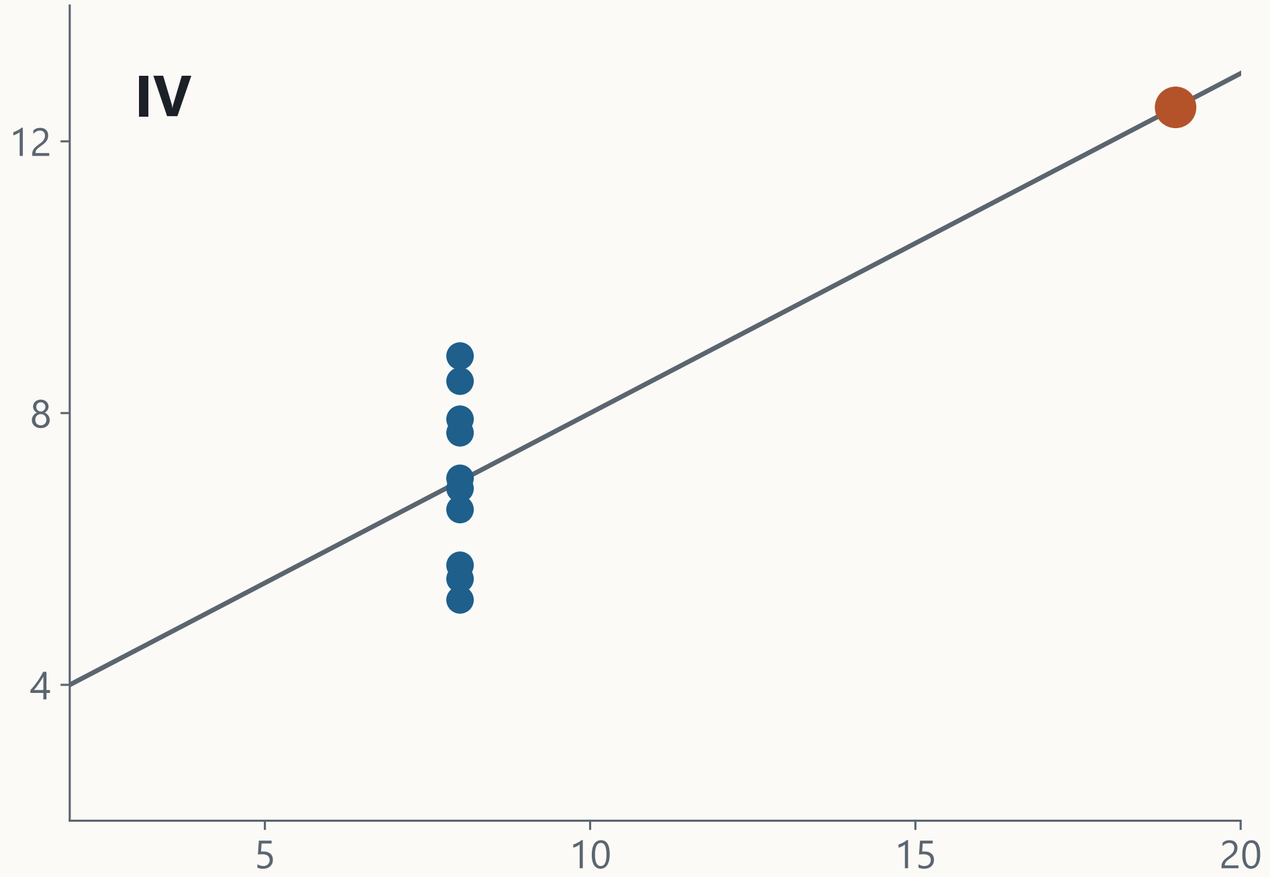


Without the highlighted point, the correlation is **1.000** and the line becomes y = 4.01 + 0.345 x.

One outlier is enough to change the slope we report.

Source: Anscombe (1973); fit without the outlier, own computation.

In IV there is no relation: a single point manufactures the whole correlation



Ten of the eleven points share the same x. The only different one decides the whole line.

Source: Anscombe (1973).

Before reporting a metric, plot the data and the residuals

- **Plot the distribution,** not only the mean: points or boxes per group (Tukey).
- **Plot the residuals** of every fit: that is where II's curve and III's outlier show up.
- **Look at every point that dominates** the result before you trust the number.

Source: Anscombe (1973); Wickham & Stryjewski, "40 years of boxplots" (2011).

APPENDIX

Sources

- Anscombe, F. J. (1973). Graphs in Statistical Analysis. The American Statistician 27(1), 17–21. DOI 10.1080/00031305.1973.10478966.
- Wickham, H. & Stryjewski, L. (2011). 40 years of boxplots. https://vita.had.co.nz/papers/boxplots.pdf
- Data and computations: figuras/hacer_figuras.py (checks the statistics before plotting).